Yuchen Miao

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Education

Northeastern University (985) *2023 – 2027 (Expected)*

B.Eng. in Communication Engineering, Sydney Smart Technology College *GPA: 3.89; Average: 88.9/100*

IELTS: Overall Band 7.0 (Listening 7.5; Speaking 7.5; Reading 6.0; Writing 6.0)

Research Experience

Hebei Key Laboratory of Marine Perception Network and Data Processing *2025 – Present*

Undergraduate Researcher

- Conduct research on explainable multimodal recommendation and fake-content detection using LLM reasoning, retrieval augmentation, graph learning, flow matching, and mixture-of-experts models.

Explainable Recommendation and Fake-Content Detection for New Media Security *2025 – Present*

National College Students' Innovation and Entrepreneurship Training Program

- Project No. 202619145026; investigated retrieval-augmented recommendation and multimodal fake-content detection for new-media security.
- Produced two peer-reviewed conference papers: "R3-REC" (ICASSP 2026) and "IMEX-FND" (WISE 2026).

Publication

1. "The '10th Juror': Open-Set Standpoint Screening for Bureaucratic Bias Detection."
   Yuchen Miao, Zijun Wang, Chang Han, Yurui Shi, Mingtai Zhang, and Siyang Xu. *EMNLP 2026 (Main Conference)*. **[CORE A*; CCF-B]**

   Introduces an open-set, standpoint-aware multi-agent framework that retrieves legal evidence, detects previously unseen bias targets, and verifies corrective rewrites for reliable bias governance in government documents. The verification step checks that mitigation preserves the original meaning. On DGDB, it reaches 0.880 F1 while limiting unnecessary interventions to 2.5%, and recovers held-out categories with 85.1% Correct@1.

2. "Anatomy of a Decision: Uncertainty-aware Hierarchical Intent Learning via Flow Matching for Multimodal Recommendation."
   Yuchen Miao, Zijun Wang, Ke Liu, and Siyang Xu. *WISE 2026*. **[CORE B; CCF-B]**

   Uses conditional flow matching to quantify visual and textual uncertainty, then constructs personalized intent hierarchies whose granularity adapts to each user's decision certainty. This enables individualized coarse-to-fine preference modeling.

3. "IMEX-FND: A Traceable Interaction-Aware Mixture-of-Experts Framework for Multimodal Fake News Detection."
   Yuchen Miao, Zijun Wang, Ke Liu, Peixuan Wang, and Chang Han. *WISE 2026*. **[CORE B; CCF-B]**

   Develops an adaptive mixture-of-experts pipeline that routes multimodal evidence and decomposes text–image interactions into uniqueness, redundancy, and synergy, yielding accurate and traceable fake-news decisions. Both routing and interaction weights remain inspectable.

4. "R3-REC: Reasoning-Driven Recommendation via Retrieval-Augmented LLMs over Multi-Granular Interest Signals."
   Yuchen Miao, Mingxuan Cui, Yitong Zhu, Yu Wang, and Siyang Xu. *ICASSP 2026*. **[CORE B; CCF-B]**

   Combines long- and short-term interest mining, similar-user retrieval, item semantic extraction, and prompt-guided reasoning to improve sequential recommendation under sparse and noisy evidence. The reasoning path makes user–item matching more transparent.

5. "MOTIF: Motivation-guided Topology Inference for Cold-start Multimodal Recommendation."
   Yurui Shi*, **Yuchen Miao***, Ximing Hu, Zijun Wang, and Chang Han. *WISE 2026*. **[CORE B; CCF-B]** *Equal contribution.

   Employs offline LLM reasoning to infer user and item motivations, reconstruct cold-start graph topology, and align semantic and structural representations without adding online LLM inference. The resulting pipeline is practical for cold-start deployment.

Preprints

1. "Universal Refinement without Interaction: Order-Optimal 1-Bit Mean Estimation."
   Yuchen Miao. *arXiv preprint*, 2026. arXiv:2607.24358.

   Constructs fully non-adaptive public-coin protocols for 1-bit mean estimation under finite central moments, achieving order-optimal refinement rates through decoder-side dyadic and continuous-scale schemes. The result shows that interaction is unnecessary in this setting.

2. “Sharp Root Anti-Concentration via Projective Incidence and Ordered Root Laws.”
Zijun Wang, **Yuchen Miao**, Yifan Hu, and Huanmin Liu. *arXiv preprint*, 2026. arXiv:2608.01670.

Establishes sharp, dimension-free root anti-concentration characterizations using projective incidence geometry and ordered root laws, with applications to regret guarantees in graph learning. The analysis removes earlier dimension-dependent losses.

Manuscripts Under Review

Descriptive titles are used; venue and coauthor details are withheld during double-blind review.

1. “Evidence-Trace Reasoning for Multimodal Misinformation Analysis.” **First Author**
Yuchen Miao et al. Manuscript under double-blind review, 2026.

Studies evidence-grounded multimodal misinformation analysis, generating auditable verification traces that connect verdicts to decisive textual and visual evidence. Evaluation considers both verdict accuracy and the fidelity of the supporting process.

2. “Tool-Assisted Evidence Verification for Automated Manuscript Review.” **First Author**
Yuchen Miao et al. Manuscript under double-blind review, 2026.

Develops a tool-assisted reviewing workflow that gathers and compresses external evidence, then produces rubric-aligned assessments with traceable support for each review claim. The design targets unsupported claims, citation errors, and unstable recommendations.

Patent

1. **Yuchen Miao**, Yu Wang, Gang Wang, Xin Feng, and Siyang Xu. “Explainable Multi-Domain Mixture-of-Experts Method for Multimodal Fake-Content Detection.” Chinese Invention Patent Application No. 202511816918.8. Filed Dec. 4, 2025; in substantive examination.

Industry Experience

China Development Bank, Shanxi Branch
Jun 2025 – Aug 2025
Information Technology Intern

- Supported information-system operations and testing, network security, and video-system administration.

Huajin Coking Coal Co., Ltd.
Jul 2024 – Sep 2024
Big Data Center Intern

- Conducted data analysis and mining; contributed to the design and optimization of data-processing workflows.

Beijing Kedinoyi Technology Co., Ltd.
Jul 2024 – Oct 2024
Embedded Systems Development Intern

- Developed device drivers in C and performed hardware debugging; received the Outstanding Intern Award.

Honors and Awards

2026
Finalist Award (Top 1%)
COMAP Mathematical Contest in Modeling (MCM)

2025
National Third Prize
Simuro Multi-Robot Penalty Kick
The 18th Advanced Robot and Simulation Technology Competition

2023 – 2026
University Third-Class Scholarship (awarded five times)
Northeastern University

2025
Provincial Second Prize
National College Students Market Research and Analysis Competition

2025
Provincial Second Prize
Hebei Provincial College Students “Research Hebei” Social Survey Program

Technical Skills

Programming and tools: Python, PyTorch, C, MATLAB, LaTeX
Machine learning: LLMs and retrieval-augmented generation, multimodal learning, recommender systems, graph learning, flow matching, mixture-of-experts, contrastive learning
Data and systems: data analysis and mining, experimental evaluation, embedded driver development, hardware debugging